

# João **Quintanialha**


Full-stack Dev · Backend Dev · Blockchain Dev

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## Summary


I am João Quintanilha, a 29-year-old Software Developer with an MSc in Computer and Telematics Engineering. I consider myself a person who is easily integrated into a team, with a natural talent for dealing with various assignments. I have strong analytical and problem-solving skills and am dynamic, resourceful, and proactive. I have excellent communication skills, smooth adaptability to multicultural environments, good organizational knowledge, and noteworthy strategic thinking. I love being engaged with a team and working together thoroughly towards a meaningful objective.


## Work Experience

### CERN

*Geneva, Switzerland*

Software Developer — *Apr. 2021 - Current*

At CERN, I am part of the Experimental Physics Department on a team responsible for the control systems of the Compact Muon Solenoid experiment (CMS). It contains a large hierarchy of subdetectors and subsystems, with control systems running over WinCC OA and Supervisory Control and Data Acquisition (SCADA). On a higher level, there is cmsonline, which holds many web applications used to monitor, execute operations, manage access authorization, etc., regarding all subdetectors. I am responsible for maintaining and developing several applications from cmsonline, mainly using Java with Oracle Application Development Framework (Oracle ADF). Applications are deployed in Weblogic Oracle Servers and managed through the Weblogic console. I have also migrated some components to Spring and actively contribute to the support role, resolving consumer's problems and helping clarify some doubts.

### Bosch Security Systems

*Ovar, Portugal*

Software Developer — *Jul. 2020 - Mar. 2021*

Part of the first international team (outside Germany) for Bosch Smart Home, I was responsible for integrating different external services and products with Bosch Home devices and services, providing interoperability across a new and improved IoT ecosystem at home. Developed some of the speech-to-action (such as open/close commands to shutters and door locks) currently existing on the market with Google Assistant and Alexa. Most functionalities were developed using Spring and Vertx. Made use of and contributed to a massive CI/CD using Jenkins. Followed Scrum framework vigorously.

### Ubiwhere

*Aveiro, Portugal*

Backend and Blockchain Developer — *Mar. 2018 - Jun. 2020*

At Ubiwhere I was in charge of help designing and develop decentralized architectures for smart-contract interoperable platforms. Develop dApps using Solidity, Truffle, and Web3 with HTTPS connection for Public Ethereum Clients. Produce RESTful APIs to serve different client applications using Python and Java Frameworks like Falcon and Spring. Environment virtualization of multiple Backend services, middleware, and Databases (Postgres and MySQL) using Docker tools. Data balance definition for hybrid applications, Database-Blockchain.

## Projects involved

### Cmsonline

- Develop new features to ease the process of specific tasks, like group handling and authorization management.
- Create study cases to migrate legacy software to modern frameworks like Spring.
- Contribute to bug-free any application already deployed and running in production.
- Create an online repository to manage dependencies better using Nexus and Openshift.
- Develop CI/CD pipelines for some applications while managing database connections and proxies.
- Create a single-place documentation page using Material for MkDocs and Openshift.
- Actively assigned to support role.

### Bosch Smart Home

- Enhance Fulfillment and Skill adapter services to provide speech control commands via virtual assistants.
- Develop services to report device states to Cloud services.
- Build and deploy overall service infrastructure using Docker containers, Jenkins, Artifactory, and AWS Lambda functions.
- Use different test frameworks to develop integration and system tests, such as JUnit and Mockito.
- Follow Scrum methodologies and Jira backlogs during each sprint.

### drEVen - Electrical vehicle charging solution

- Implement Steve to make use of Open Core Protocol (OCP) for Electric vehicle charging using Spring, Java.
- Develop and define Smart-Contract logic using Solidity and Web3.
- Produce RESTful APIs to serve different client applications using Falcon, Python.
- Environment virtualization of multiple Backend services, Middlewares, and Databases (Postgres and MySQL) using Docker, Azure, and Digital Ocean tools.

### Dyagora - Wine certification platform

- Close relationship with our partner, the wine producer Murganheira, to understand requirements.
- Help define Database tables to reach an interoperable Blockchain-Database relationship.
- Develop and define Smart-Contract logic using Solidity and Web3.
- Help developing parts of the partner's Dashboard using ReactJS.
- Help and teach Backend developers to interact with Ethereum clients and contracts.

### Physiotherapy management platform

- Client's requirement gathering and analysis for technology usage.
- Develop Backend service components using Falcon, Python.
- Develop a web Frontend for administrative access using ReactJS.
- Test, deploy and maintain services @ Contabo using Docker.
- Develop the Physiotherapy's website to promote their services using ReactJS.

## Education

| | | |
|---|---|---|
| 2012 - 2018 | **MSC & BSc Computer and Telematics Engineering**, University of Aveiro | *Aveiro, Portugal* |
| 2017 - 2018 | **MSc Software Engineering and Information Systems (Erasmus+)**, University of Zagreb | *Zagreb, Croatia* |
| 2016 - 2017 | **BSc Information Technologies (Erasmus+)**, University of Silesia | *Katowice, Poland* |

## Certificates & Awards

| | | |
|---|---|---|
| 2022 | **Scrum**, Professional Scrum Master I (PSM 1) | *Scrum.org* |
| 2021 | **Languages**, French Certificate A2.1 (CECRL) | *Supercomm Group* |
| 2020 | **Docker**, Docker Mastery: with Kubernetes + Swarm | *Udemy, online* |
| 2020 | **Java**, Advanced Springboot Development | *Udemy, online* |
| 2019 | **Blockchain**, Dyagora technical pitch - Blockchers H2020 funded project | *Budapest, Hungary* |
| 2019 | **Health**, Basic life support | *Aveiro, Portugal* |
| 2018 | **Blockchain**, Blockchain Development on Hyperledger Fabric using Composer | *Udemy, online* |
| 2018 | **Blockchain**, Ethereum and Solidity: Developer's Guide | *Udemy, online* |
| 2014 | **Languages**, First Certificate of English (FCE) | *Aveiro, Portugal* |

## Writing

### Blockchain for non-tech-savvy

*Medium*

Blockchain Developer — *Mar. 2020*

- A drafted article that targets non-tech profiles to explain the mechanisms of Blockchain technology through a quotidian basis.

### Multi-agent loyalty program over blockchain

*University of Aveiro*

Computer Engineer Student — *Dec. 2019*

- Master's thesis written as the conclusion for the Computer Engineering degree.

## Languages

**Portuguese** → Native speaker
**English** → First Certificate of English (FCE)
**French** → Limited working proficiency (A2.1)
**Spanish** → Limited working proficiency

## Competency Matrix

**Beginner (1)** → Academic

**Intermediate (2)** → Online and certificated training

**Advanced (3)** → Personal project experience

**Professional (4)** → Professional experience

**Expert (5)** → 5+ years of personal and professional experience

| Languages | |
|---|---|
| Java | 5 |
| Python | 4 |
| Javascript | 4 |
| Solidity | 4 |
| HTML & CSS | 3 |
| **Frameworks** | |
| Spring | 4 |
| Falcon | 4 |
| Truffle | 4 |
| NodeJS | 4 |
| Mockito | 3 |
| Django | 3 |
| ReactJS | 3 |
| Mocha | 3 |
| Flask | 2 |
| **Databases + ORMs** | |
| Postgresql | 4 |
| SQL Alchemy | 4 |
| Hibernate | 4 |
| SQL Server | 4 |
| **Others** | |
| Git | 5 |
| Linux OS (Bash) | 5 |
| Docker | 4 |
| Google & Amazon Cloud APIs | 4 |
| Jenkins | 4 |